

SOFTWARE REQUIREMENTS SPECIFICATION

Company Internal Reporting System

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Project	Internship Project Assignment

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document describes the functional and non-functional requirements for the Company Internal Reporting System. It is intended to guide the design, development, and testing of the system and serves as a reference for all project stakeholders including developers, reviewers, and approvers.

1.2 Scope

The Company Internal Reporting System is a web-based full-stack application that enables structured report submission, review, and approval within an organization. The system supports multiple user roles and enforces a two-stage approval workflow. Core capabilities include:

- User and organization management (departments, teams, employees)
- Admin-defined report scheduling and frequency settings
- Employee report submission via file upload or online forms
- Two-stage review and approval workflow (Department Reviewer → Final Approver)
- In-app and email notification system
- Report tracking, history, and status management
- Role-based dashboards and analytics

1.3 Definitions, Acronyms, and Abbreviations

Term	Definition
SRS	Software Requirements Specification
Admin	System administrator with full management privileges
Employee	A staff member who submits reports
Department Reviewer	A user assigned to review department reports at Stage 1
Final Approver	A senior officer (e.g., COO) who performs Stage 2 approval
DFD	Data Flow Diagram
ERD	Entity Relationship Diagram
API	Application Programming Interface
UI	User Interface
RBAC	Role-Based Access Control

1.4 References

- Internship Project Assignment Document – Mugisha Yves
- Next.js Official Documentation – nextjs.org

- Node.js & Express Documentation – nodejs.org / expressjs.com
- PostgreSQL / MySQL Documentation

1.5 Overview

The remainder of this document is organized as follows:

1. Section 2 – System Overview: High-level description of the system and its context.
2. Section 3 – User Roles and Permissions: Roles defined and their access levels.
3. Section 4 – Functional Requirements: Detailed features the system must perform.
4. Section 5 – Non-Functional Requirements: Quality attributes and constraints.
5. Section 6 – System Modules: Module breakdown.
6. Section 7 – Use Cases: Actor-based use case descriptions.

2. System Overview

2.1 System Description

The Company Internal Reporting System is a centralized platform designed to streamline how employees submit, track, and get their reports reviewed and approved. The system removes manual, paper-based processes by providing a digital workflow with notifications, role-based access, and full audit trails.

2.2 System Context

The system operates within a company's internal network and is accessible via web browsers. It interfaces with an email server for notifications and stores all data in a relational database. The primary actors interacting with the system are:

- Admin – manages organizational structure and report schedules
- Employees – submit periodic reports
- Department Reviewers – conduct first-stage reviews
- Final Approver – provides final approval or rejection

2.3 System Architecture Overview

The system follows a three-tier architecture:

- Frontend: Next.js (React-based) — delivers responsive, role-based UI
- Backend: Node.js with Express — RESTful API layer with business logic
- Database: PostgreSQL or MySQL — stores all structured data

Communication between frontend and backend occurs via REST APIs. Authentication is managed using JWT (JSON Web Tokens). Email notifications are sent via an SMTP service.

2.4 Assumptions and Constraints

- All users must have a valid company email address to register
- The system is intended for internal use only and is not publicly exposed
- Internet access is required for email notifications
- Users access the system via modern web browsers (Chrome, Firefox, Edge)
- Maximum file upload size for report attachments is 10MB

3. User Roles and Permissions

3.1 Role Definitions

The system defines four distinct user roles, each with specific permissions:

Role	Description	Key Permissions
Admin	System administrator	Full access: manage users, departments, teams, report types, schedules, and system configuration
Employee	Staff member who submits reports	Submit reports, view own submissions, track status, receive notifications
Department Reviewer	First-stage reviewer per department	View and review department reports, approve/reject/request changes at Stage 1
Final Approver (COO)	Final decision maker	View all reports after Stage 1, perform final approval/rejection, view analytics

3.2 Permission Matrix

Feature	Admin	Employee	Dept. Reviewer	Final Approver
Create Departments/Teams	Yes	No	No	No
Register Employees	Yes	No	No	No
Define Report Schedules	Yes	No	No	No
Submit Reports	No	Yes	No	No
Stage 1 Review	No	No	Yes	No
Stage 2 Final Approval	No	No	No	Yes
View All Reports	Yes	No	No	Yes
View Own Reports	Yes	Yes	Yes	Yes
Admin Dashboard	Yes	No	No	No
Department Dashboard	Yes	No	Yes	No
Reviewer Dashboard	No	No	Yes	No
Analytics & Compliance	Yes	No	No	Yes

4. Functional Requirements

4.1 User and Organization Management

FR-01: Department Management

- Admin shall be able to create, update, and delete departments
- Each department shall have a unique name and optional description
- Admin shall assign a Department Reviewer to each department

FR-02: Team Management

- Admin shall be able to create teams within departments
- Teams shall belong to exactly one department
- Admin shall assign employees to teams

FR-03: Employee Registration

- Admin shall register employees by providing: name, email, role, department, and team
- Registered employees shall receive a system-generated password via email
- Employees shall be able to update their own profile and password

4.2 Report Scheduling

FR-04: Schedule Definition

- Admin shall define report frequency: monthly, bi-weekly, weekly, or custom
- Admin shall assign report types to specific departments or teams
- System shall automatically generate reporting timelines based on defined schedules

FR-05: Automatic Reminders

- System shall notify employees when a report submission is due
- Reminders shall be sent 3 days and 1 day before the deadline

4.3 Report Submission

FR-06: Report Submission

- Employees shall submit reports via an online form or file upload (PDF, DOCX, XLSX)
- Employees shall only see reports assigned to their department/team/schedule
- Employees shall be able to view the status of all their submissions
- Late submissions shall be marked and flagged on the dashboard

4.4 Review and Approval Workflow

FR-07: Stage 1 – Department Review

- Each submitted report shall be routed to the assigned Department Reviewer

- Reviewer shall be able to: Approve (sends to Stage 2), Reject (returns to employee), Request Changes (employee must resubmit)
- Reviewer shall be able to add comments with any action

FR-08: Stage 2 – Final Approval

- Reports approved at Stage 1 shall be forwarded to the Final Approver (COO)
- Final Approver shall be able to: Approve, Reject, or Request Changes
- All actions shall be logged with timestamps and comments

4.5 Notification System

FR-09: Notification Triggers

Event	Notified Users
Report is requested / due	Employee
Report is submitted	Department Reviewer, Admin
Report is reviewed (Stage 1)	Employee, Final Approver
Report is approved (Stage 2)	Employee, Admin
Report is rejected	Employee
Changes requested	Employee

FR-10: Notification Channels

- In-app alerts — displayed in the notification bell icon
- Email notifications — sent to registered email address

4.6 Report Tracking and Records

FR-11: Status Tracking

- Each report shall have a status: Pending, Submitted, Under Review, Changes Requested, Approved, Rejected
- System shall maintain a full status history with timestamps for each report
- Reports shall be filterable by department, team, date range, type, and status

4.7 Dashboard and Analytics

FR-12: Admin Dashboard

- Overview of total reports, pending approvals, compliance rates
- Reports breakdown by department and status

FR-13: Department Dashboard

- Submission progress for the current reporting period
- List of pending and completed reports within the department

FR-14: Reviewer Dashboard

- List of reports awaiting Stage 1 review
- Quick-action buttons for approve/reject/request changes

FR-15: Search and Filter

- All report lists shall support filtering by: Department, Team, Date, Report Type, Status
- Search by employee name or report title

5. Non-Functional Requirements

ID	Category	Requirement
NFR-01	Performance	The system shall load any page within 3 seconds under normal load (up to 100 concurrent users).
NFR-02	Security	All API endpoints shall require JWT authentication. Passwords shall be hashed using bcrypt.
NFR-03	Security	Role-based access control (RBAC) shall be enforced on both frontend and backend.
NFR-04	Availability	The system shall maintain 99% uptime during working hours (8 AM – 6 PM).
NFR-05	Usability	The UI shall be responsive and fully functional on desktop and mobile browsers.
NFR-06	Scalability	The database schema and API design shall support future addition of new departments, roles, and report types without major refactoring.
NFR-07	Maintainability	Code shall follow clean architecture principles. All API endpoints shall be documented.
NFR-08	Data Integrity	All database transactions shall be ACID-compliant. Foreign key constraints shall be enforced.
NFR-09	Auditability	Every action on a report (submission, review, approval, rejection) shall be logged with user ID and timestamp.
NFR-10	Reliability	The system shall gracefully handle errors and display user-friendly error messages without exposing internal details.

6. System Modules

The system is divided into the following logical modules:

Module	Description	Key Components
Authentication Module	Handles login, logout, password management, and JWT token issuance/validation.	Login page, JWT middleware, password reset flow
User Management Module	Admin functionality for managing employees, roles, departments, and teams.	User CRUD, role assignment, department/team management
Report Scheduling Module	Admin defines and manages report frequencies, types, and deadlines.	Schedule configuration, timeline auto-generation, deadline tracker
Report Submission Module	Employee-facing module for submitting and tracking reports.	Submission form, file uploader, status tracker
Review & Approval Module	Manages the two-stage approval workflow.	Stage 1 reviewer interface, Stage 2 approver interface, comment and action logging
Notification Module	Triggers and dispatches in-app and email notifications.	Notification service, email integration (SMTP), notification bell UI
Dashboard & Analytics Module	Role-specific dashboards with report overviews and compliance metrics.	Admin/reviewer/employee dashboards, charts, compliance reports
Audit & History Module	Stores and displays complete status history for every report.	Status log table, timeline view, filter/search

7. Use Cases

7.1 Use Case List

Use Case ID	Use Case Name	Primary Actor
UC-01	Register and Manage Employees	Admin
UC-02	Create Departments and Teams	Admin
UC-03	Define Report Schedule	Admin
UC-04	Submit a Report	Employee
UC-05	Track Report Status	Employee
UC-06	Perform Stage 1 Review	Department Reviewer
UC-07	Perform Stage 2 Final Approval	Final Approver
UC-08	Receive and View Notifications	All Users
UC-09	View Dashboard	All Users
UC-10	Search and Filter Reports	Admin, Reviewer, Final Approver

7.2 Use Case Descriptions

UC-04: Submit a Report

Field	Detail
Use Case ID	UC-04
Name	Submit a Report
Actor	Employee
Precondition	Employee is logged in and has a report due based on the assigned schedule.
Trigger	Employee navigates to the 'My Reports' section and selects a pending report.
Main Flow	1. Employee opens the report submission form. 2. Employee fills in the report fields or uploads a document. 3. Employee clicks 'Submit'. 4. System validates the submission. 5. System saves the report with status 'Submitted'. 6. System notifies the Department Reviewer.
Alternate Flow	If validation fails, system displays errors and the employee corrects them.
Postcondition	Report is stored with status 'Submitted' and the Department Reviewer is notified.
Exception	If submission deadline has passed, report is flagged as late but submission is still allowed.

UC-06: Perform Stage 1 Review

Field	Detail
Use Case ID	UC-06
Name	Perform Stage 1 Review
Actor	Department Reviewer
Precondition	A submitted report exists in the reviewer's queue.
Trigger	Reviewer opens the Reviewer Dashboard and selects a pending report.
Main Flow	1. Reviewer opens the report and reads the content. 2. Reviewer adds optional comments. 3. Reviewer selects action: Approve, Reject, or Request Changes. 4. System updates report status accordingly. 5. System notifies the Employee and, if approved, the Final Approver.
Alternate Flow	Reviewer requests changes; employee receives notification and must resubmit.
Postcondition	Report status is updated and relevant parties are notified.
Exception	If no action is taken within the deadline, Admin receives an alert about pending review.

UC-07: Perform Stage 2 Final Approval

Field	Detail
Use Case ID	UC-07
Name	Perform Stage 2 Final Approval
Actor	Final Approver (COO)
Precondition	Report has been approved at Stage 1 and forwarded to the Final Approver.
Trigger	Final Approver receives in-app notification or email about a report awaiting approval.
Main Flow	1. Final Approver navigates to the dashboard and opens the report. 2. Final Approver reviews the report and optional Stage 1 comments. 3. Final Approver selects: Approve or Reject. 4. System updates report status to 'Approved' or 'Rejected'. 5. System notifies the Employee and Admin.
Alternate Flow	Final Approver requests changes; report returns to employee for resubmission.
Postcondition	Report is fully resolved with a final status. All parties are notified.
Exception	If report is rejected, it must include a mandatory reason/comment.

7.3 System Use Case Diagram Summary

Below is a tabular representation of actors and use cases:

Actor	Use Cases
Admin	UC-01, UC-02, UC-03, UC-09, UC-10
Employee	UC-04, UC-05, UC-08, UC-09
Department Reviewer	UC-06, UC-08, UC-09, UC-10
Final Approver	UC-07, UC-08, UC-09, UC-10